Faculty of Science, Technology, Engineering and Mathematics
M303 Further pure mathematics

M303

TMA 03 2025J

Mainly covers Book C *Numbers and rings* **Cut-off date 27 January 2026**

You will find instructions for completing TMAs in the Assessment resources area of the M303 website. Please remind yourself of these instructions before beginning work on each TMA. You can submit each TMA electronically by using the University's online TMA/EMA service or by post together with completed TMA form (PT3). In both cases the University must receive it by the cut-off dated given above unless you have agreed an extension in advance with your tutor.

Half of each TMA is formative and the other half summative. The formative questions are designed to *extend* your understanding of the topics, while the summative ones place more emphasis on *assessing* your understanding of the topics. Both types of question will be marked by your tutor, but (unlike the summative questions) marks for the formative questions do not count towards your final grade. You should submit your solutions to both sets of questions to your tutor by the cut-off date.

The marks allocated to each part of a question are indicated in the margin.

Although many of the questions on these assignments require numerical answers, the questions invariably carry *method marks*.

The words listed below, when used in questions, should be interpreted as indicated.

prove, show, explain, justify	Clear reasoning and explanation for all steps are called for.
determine, find, devise, calculate, compute	An indication of the method used and all working in arriving at an answer should be given.
deduce	Clear explanation of how one result follows from another is required.
solve	Working must be shown. A numerical answer alone is not sufficient.
evaluate, give, write down, list	Answer suffices. No explanation need be given.
hence	No marks will be awarded for any alternative method.

<u>PLAGIARISM WARNING</u> – the use of assessment help services and websites

The **work that you submit for any assessment/examination on any module should be your own**. Submitting work produced by or with another person, or a web service or an automated system, **as if it is your own** is cheating. It is **strictly forbidden** by the University.

You should not:

- provide any assessment question to a website, online service, social media platform or any individual or organisation, as this is an infringement of copyright
- request answers or solutions to an assessment question on any website, via an online service or social media platform, or from any individual or organisation
- use an automated system (other than one prescribed by the module) to obtain answers or solutions to an assessment question and submit the output as your own work
- discuss examination questions with any other person, including your tutor.

The University actively monitors websites, online services and social media platforms for answers and solutions to assessment questions, and for assessment questions posted by students. Work submitted by students for assessment is also monitored for plagiarism.

A student who is found to have posted a question or answer to a website, online service or social media platform and/or to have used any resulting, or otherwise obtained, output as if it is their own work has committed a disciplinary offence under our Code of Practice for Student Discipline. **This means the academic reputation and integrity of the University has been undermined.**

The Open University's Academic Conduct Policy defines plagiarism in part as:

- using text obtained from assignment writing sites, organisations or private individuals
- obtaining work from other sources and submitting it as your own.

If it is found that you have used the services of a website, online service or social media platform, or that you have otherwise obtained the work you submit from another person, this is considered serious academic misconduct and you will be referred to the Central Disciplinary Committee for investigation.

This TMA is based on Book C (Numbers and rings), except for the formative Question 8, which covers the start of Book D.

Part A is **summative**. These questions assess your knowledge of the module. You must not discuss these questions in the forums, and your tutor is not allowed to help you directly.

Part B is **formative**. You should choose (at most) two formative questions to submit to your tutor for marking.

Doing the formative questions will help you to understand key concepts covered in the text. You are encouraged to discuss these questions in the module forums and with your tutor (who is allowed to help you).

Question 8 is designed to help you with TMA 04. Questions 9 and 10 cover material that is more central to the module, while Question 11 is designed to expand your understanding beyond the core material.

Your tutor will mark all the summative answers that you submit and the first two of any formative answers that you submit. If you have attempted at least one formative question then when your TMA is returned you will be sent worked solutions to all the formative questions. Note that only your marks for Part A count towards your overall continuous assessment score.

Part A Summative

This TMA uses polynomials. You may assume that two polynomials, $a_0 + a_1x + \cdots + a_nx^n$ and $b_0 + b_1x + \cdots b_mx^m$, with $m, n \geq 0$, are equal if and only if $n = m$ and $a_i = b_i$ for all i (this is sometimes known as 'equating coefficients' or 'comparing coefficients').

Question 1 – 7 marks

This M303 question tests your understanding of material covered in Chapter 9 (and Chapter 2). You should not use a calculator for this question and you should show your workings.

(a) Determine the smallest positive integer that has exactly $1573 = 11^2 \cdot 13$ distinct positive divisors, including itself and 1. You should leave your answer in the form $p_1^{q_1} \cdots p_r^{q_r}$. *[4]*

(b) Let n be a positive integer such that $7n + \sigma(n)$ is divisible by 8 (where $\sigma(n)$ denotes the sum of the positive divisors of n).

Either prove the following or find a counterexample.

If $n = p^2$ where p is prime, then $p \equiv 7 \pmod{8}$. *[3]*

Question 2 – 8 marks

This M303 question tests your understanding of material covered in Chapter 10. You should not use a calculator for this question.

(a) Determine whether or not the quadratic congruence

$$2x^2 + 3x + 4 \equiv 0 \pmod{7}$$

has solutions:

(i) by using Euler's Criterion; *[2]*

(ii) by using Gauss's Lemma. *[2]*

(b) Use the Law of Quadratic Reciprocity (at least once somewhere in this question) and other results such as Theorem 2.2, Proposition 3.2 and Proposition 4.7 of Chapter 10, to determine the values of the Legendre symbols

$$(-48/229) \text{ and } (151/43).$$

You must precisely reference the M303 results that you use (by, for example, saying 'Thm 2.2(a) Chap 10', and not 'Thm 2.2 Chap 10' if you need part (a) of Theorem 2.2).

(Note that 229 and 43 are prime.) *[4]*

Question 3 – 7 marks

This M303 question tests your understanding of material covered in Chapter 11. You must show all workings for this question.

Find the hcf in $\mathbb{Q}[x]$ of

$$f(x) = 3x^5 + 11x^4 + 4x^3 - 8x^2 - 6x$$

and

$$g(x) = 3x^3 + 11x^2 + 7x + 3.$$

[7]

Question 4 – 6 marks

This M303 question tests your understanding of material covered in Chapter 11.

Let $f(x) = x^6 + 4x^3 - 2x^2 + 6x + 2$.

(a) Use the Rational Root Test to show that $f(x)$ has no rational roots. [4]

(b) Use Eisenstein's Criterion to show (again) that $f(x)$ has no rational roots. [2]

Question 5 – 10 marks

This M303 question tests your understanding of material covered in Chapter 12.

(a) (i) Suppose that $\sqrt{50} = \dfrac{m}{n}$, where m and n are positive integers. Show that

$$\sqrt{50} = \frac{50n - 7m}{m - 7n}.$$

[2]

(ii) Explain how the equation shown in (a)(i) and the method of infinite descent can be used to deduce that $\sqrt{50}$ is irrational. [3]

(b) (i) Write 578 as the sum of two squares in two different ways, showing all workings where relevant. [3]

(ii) Can 1715 be written as the sum of two squares? Justify your answer. [2]

Question 6 – 10 marks

This M303 question tests your understanding of material covered in Chapter 11 and Chapter 12.

Let

$$R = \{a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_nx^n \,:\, a_0, a_1, \ldots, a_n \in \mathbb{Z}_{59}, a_1 = 0, n \geq 0\}$$

be the set of all polynomials with coefficients in $\mathbb{Z}_{59}$ such that the coefficient of x is zero.

(a) Show that R is a subring of $\mathbb{Z}_{59}[x]$. What are the units of R? [4]

(b) Prove that x^2 is an irreducible element of R. [3]

(c) Show that x^2 does not divide x^3 in R, and hence (or otherwise) prove that x^2 is not a prime element of R. *Hint:* Consider the element x^6. [3]

Question 7 – 2 marks

For this TMA, we would like you to post something in the Book C forum that is about the mathematics in Book C. To earn full marks, provide either a screenshot of your forum post or directions as to where to find it (giving the thread title, time and date). (If you are not in a position to provide such evidence, contact your tutor for advice.) *[2]*

Part B Formative

Question 8 – 11 marks

This question concerns Chapter 13 of Book D, and relates to Question 5 in TMA 04.

We define three subsets of the plane (using set notation) as follows:

$$A = \{(x, y) \in \mathbb{R}^2 : 1 < x^2 + y^2 \leq 5\},$$
$$B = \{(x, y) \in \mathbb{R}^2 : y \geq 1\}$$

and

$$C = A \cap B.$$

(a) Sketch on separate diagrams, by hand or otherwise, the sets A, B and C, taking care to distinguish between points that lie inside and outside the sets. [6]

When sketching these sets, take care to indicate which points are included: use dashed lines to indicate boundaries that are not included and solid lines to indicate boundaries that are. Use small open circles ∘ *to indicate any 'corner' points on a boundary that are not included and small closed circles* • *to indicate any 'corner' points that are included.*

(b) Use Definition 3.4 of Chapter 13 to determine whether the limit of the sequence $\left((0, 1 + \frac{1}{n})\right)_{n=1}^{\infty}$ is in C. [5]

Question 9 – 14 marks

This question gives you practice in working with polynomials. It also gives you a chance to practise use of the $\sum$ *notation for summation.*

(a) Let $f(x)$ be a polynomial in $\mathbb{Q}[x]$ with

$$f(x) = a_0 + a_1 x + \cdots + a_{n-1}x^{n-1} + a_n x^n = \sum_{i=0}^{n} a_i x^i.$$

Write down each of the following products using the summation notation, taking care in your answer to ensure that the summands each contain only a single power of x.

(i) $x\, f(x)$

(ii) $(x + 1)\, f(x)$

(iii) $(b_0 + b_1 x + b_2 x^2)\, f(x)$ [4]

Note: it is a convention in summation notation that if the sum extends to a range for which summands are not defined, then the undefined summands are taken to be zero. So if $a_1 = 1$, $a_2 = 2$ and $a_3 = 3$ are the only defined summands, then $\sum_{i=0}^{4} a_i = 0 + 1 + 2 + 3 + 0 = 6$.

(b) Verify that $\mathbb{Z}_7[x]$ satisfies the ring axiom R6:
if $f(x) = \sum_{i=0}^{n} a_i x^i \in \mathbb{Z}_7[x]$ and $g(x) = \sum_{j=0}^{m} b_j x^j \in \mathbb{Z}_7[x]$, where $m, n \geq 0$ are integers, then $f \cdot g \in \mathbb{Z}_7[x]$. [4]

(c) (i) Write down the units in $\mathbb{Z}_7[x]$. [2]

(ii) Find the unique monic polynomial that is an associate of 4 in $\mathbb{Z}_7[x]$. [2]

(iii) Find the unique monic polynomial that is an associate of $5x^2 + 3$ in $\mathbb{Z}_7[x]$. [2]

Question 10 – 11 marks

In this question, you will gain practice in testing the axioms for rings and fields, and help in understanding how to manipulate general algebraic expressions.

Let $R = (\mathbb{Z}, \oplus, \odot)$ denote the set of integers equipped with the operations $\oplus$ and $\odot$ defined as follows. For $a, b \in \mathbb{Z}$,

$$a \oplus b = a + b + 1,$$
$$a \odot b = ab + a + b.$$

(a) Prove that $R = (\mathbb{Z}, \oplus, \odot)$ is a commutative ring. [*5*]

(b) Is R an integral domain? Justify your answer. [*3*]

(c) Is R a field? Justify your answer. [*3*]

Question 11 – 14 marks

This question asks you to use norms, irreducibles and primes to find a family of rings $\mathbb{Z}[\sqrt{-n}]$ *that are not UFDs.*

Let $n > 2$ be an even integer that is not divisible by 4.

(a) Show that the norm $N(a + b\sqrt{-n}) = a^2 + nb^2$ is a multiplicative norm for $\mathbb{Z}[\sqrt{-n}]$. [*2*]

(b) Show that if u is a unit in $\mathbb{Z}[\sqrt{-n}]$, then $N(u) = 1$, and hence find all the units in $\mathbb{Z}[\sqrt{-n}]$. [*3*]

(c) Prove that the element 2 is irreducible in $\mathbb{Z}[\sqrt{-n}]$. [*5*]

(d) Show that 2 does not divide $\sqrt{-n}$. [*1*]

(e) By showing that 2 is not prime in $\mathbb{Z}[\sqrt{-n}]$, prove that $\mathbb{Z}[\sqrt{-n}]$ is not a UFD. [*3*]